

Q1 [12 Points]

For each problem below, determine whether it is a linear program. Briefly justify every answer.

(a) [3 Points]

$$\begin{array}{ll} \min & x_1 + x_2 \\ \text{s.t.} & x_1 + 2x_2 < 5, \\ & x_1, x_2 \geq 0. \end{array}$$

(b) [3 Points]

$$\begin{array}{ll} \min & x + y \\ \text{s.t.} & x \sin \theta + y \cos \theta \leq 1 \quad \text{for all } \theta \in [0, 2\pi], \\ & x, y \in \mathbb{R}. \end{array}$$

(c) [3 Points]

$$\begin{array}{ll} \max & x_1 + x_2 \\ \text{s.t.} & x_1 x_2 \leq 1, \\ & x_1, x_2 \geq 0. \end{array}$$

(d) [3 Points]

$$\begin{array}{ll} \max & 3x_1 - 2x_2 \\ \text{s.t.} & x_1 + 4x_2 \leq 7, \\ & -x_1 + x_2 = 1, \\ & x_1, x_2 \geq 0. \end{array}$$

Q2 [5 Points] Let $\{v_1, v_2, \dots, v_m\} \subseteq \mathbb{R}^n$ be a linearly independent subset of vectors. Suppose $u \in \mathbb{R}^n$ is not in the span of $\{v_1, v_2, \dots, v_m\}$. Prove that the set $\{v_1, v_2, \dots, v_m, u\}$ is linearly independent.

Q3 [10 Points] **Grand River Cider Company** has a processing plant in each of Kitchener, Cambridge, Stratford, and London. The company has produced 800 litres of a promotional cider at its Kitchener plant. Every litre of this cider that reaches London can be sold for 70 cents.

The cider may be transported through the directed pipeline network shown in Figure 1. The first number displayed beside each arc is the transportation cost, in cents per litre. The second number is the maximum capacity of that pipeline, in litres.

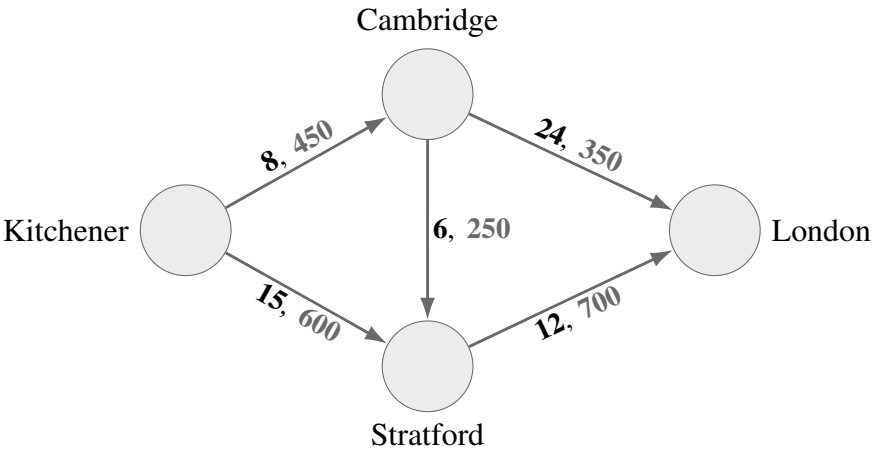


Figure 1: The directed pipeline network. Arc labels give **cost in cents per litre** followed by **capacity in litres**.

Cambridge and Stratford are transshipment plants: they neither produce nor consume any of the promotional cider. Assume that no cider is lost while it is transported. The company is not required to transport all 800 litres from Kitchener.

Formulate a linear program that determines the maximum profit the company can earn from this promotion. Do not solve the linear program. Clearly define all decision variables.

Solutions

Q1 (a) The problem is **not a linear program as written**, because the constraint

$$x_1 + 2x_2 < 5$$

is a strict inequality.

(b) The problem is **not a linear program** since

$$x \sin \theta + y \cos \theta \leq 1 \quad \text{for all } \theta \in [0, 2\pi]$$

contains infinitely many constraints.

(c) The problem is **not a linear program**. The constraint

$$x_1 x_2 \leq 1$$

contains the product of two decision variables, which is a nonlinear constraint.

(d) The problem is **a linear program**. The objective function is linear, and all its constraints are linear equalities or non-strict linear inequalities. It also has finitely many variables and constraints.

Q2 Read the solution of A1 Problem 4 (c).

Q3 Read the solution of A1 Problem 3. The LP formulation for this variant is given below.

For every directed pipeline (i, j) , define

$$x_{ij} = \text{number of litres of promotional cider transported from plant } i \text{ to plant } j.$$

In particular, the decision variables are

$$x_{KC}, \quad x_{KS}, \quad x_{CS}, \quad x_{CL}, \quad x_{SL},$$

where K , C , S , and L denote Kitchener, Cambridge, Stratford, and London, respectively. With similar argument as in Problem 3 of A1, we obtain

$$\begin{aligned} \max \quad & 70(x_{CL} + x_{SL}) - 8x_{KC} - 15x_{KS} - 6x_{CS} - 24x_{CL} - 12x_{SL} \\ \text{s.t.} \quad & x_{KC} + x_{KS} \leq 800 \\ & x_{KC} - x_{CS} - x_{CL} = 0 \\ & x_{KS} + x_{CS} - x_{SL} = 0 \\ & 0 \leq x_{KC} \leq 450 \\ & 0 \leq x_{KS} \leq 600 \\ & 0 \leq x_{CS} \leq 250 \\ & 0 \leq x_{CL} \leq 350 \\ & 0 \leq x_{SL} \leq 700 \end{aligned}$$